4 A thermocouple is a device that is used as a thermometer.

(a) Fig. 4.1 shows a beaker that contains molten sulfur at an initial temperature greater than 400 °C.

(i) On Fig. 4.1, sketch and label a diagram of a thermocouple that is used to determine the temperature of the sulfur as it cools to room temperature.

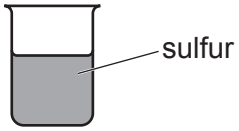


Fig. 4.1

[4]

(ii) Describe briefly how the temperature of the sulfur in the beaker is deduced.

.....

.....

..... [2]

(b) State **one** advantage of using a thermocouple to measure temperature rather than using a liquid-in-glass thermometer.

.....

..... [1]

[Total: 7]